A Systems Chronomechanobiology Model of the Intervertebral Disc

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Introduction

The intervertebral disc (IVD) is an avascular load-bearing organ sensitive to mechanotransduction (MT) and nutrient transport phenomena. It possesses an autonomous circadian clock (CR) entrained by local mechanical cues [1]. Altered mechanical loading disrupts circadian rhythmicity, while CR disruption promotes degenerative phenotypes, suggesting a functional interplay between MT and CR regulation [2]. However, the underlying molecular mechanisms are poorly understood. To contribute to fill this gap, we develop a systems-level model integrating mechanical loading, nutrient transport, MT and CR signalling to investigate the coordinated regulation of IVD homeostasis and degeneration.

Method

A multiscale computational framework based on a coupled system of ordinary differential equations (ODEs) was developed to integrate mechanical loading, MT and CR regulation (Fig. 1). Daily mechanical loading was prescribed as a time-dependent function, L(t), representing six lifestyle-associated loading conditions (normal; sedentary; exercise; microgravity; heavy load; nightshift). For each condition, L(t) was a sinusoidal approximation of the diurnal loading cycle, with condition-specific minimum and maximum loads, phase shift, and loading duration parameterized from published in vivo measurements.

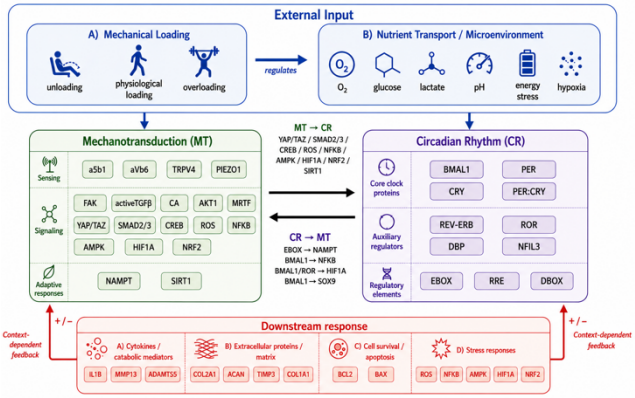


Fig. 1: Schematic of the MT–CR modelling framework.

Mechanical loading influenced the model through two complementary pathways: (i) direct MT, where L(t) activates the mechanosensors α5β1, αVβ6, TRPV4 and PIEZO1 (L(t)→Mechanosensors→MT), and (ii) indirect MT, where loading regulates nutrient transport, determining oxygen, glucose, lactate, pH and cellular energy status that subsequently modulate MT (L(t)→Nutrient transport→[O2, Glu, Lac, pH, Energy]→MT). Both pathways converge within a coupled MT–CR network to predict tissue homeostasis, matrix remodelling and cell-fate responses under different lifestyle-associated loading conditions.

Results

Lifestyle-specific mechanical loading established distinct mechanotransductive and metabolic environments (Fig. 2A), differentially regulating circadian dynamics and downstream metabolism. Exercise-enriched loading produced the largest BMAL1 oscillation amplitude (0.431), whereas physiological, sedentary, excessive loading and microgravity reduced BMAL1 amplitudes to 0.404, 0.355, 0.335 and 0.310, respectively. BMAL1 peak timing shifted from 20.1 h under physiological loading to 22.2 h during microgravity unloading and 23.3 h under nightshift loading, while the oscillation period remained close to 24 h (24.0–24.4 h) across all conditions (Fig. 2C). Circadian changes triggered coordinated metabolic responses. Exercise-enriched loading preserved anabolic matrix synthesis (COL2A1 = 0.599), whereas sedentary loading (0.586), excessive loading (0.530) and microgravity unloading (0.279) progressively reduced COL2A1 expression (Fig. 2B). Conversely, MMP13 increased from 0.027 under exercise-enriched loading and 0.041 under physiological loading to 0.088, 0.122 and 0.552 under sedentary, excessive loading and microgravity unloading, respectively (Fig. 2D).

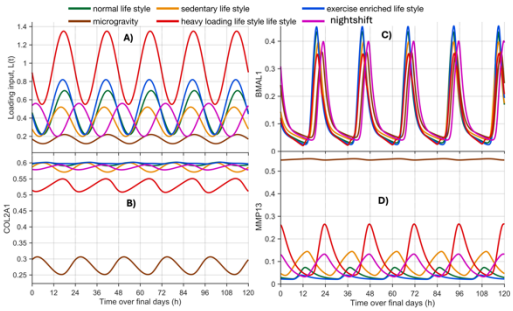


Fig. 2: Nodal responses to lifestyle-specific mechanical loading.

Discussion

The model reproduced experimentally measured load-dependent circadian adaptations, including reduced BMAL1 amplitude, phase delay and preservation of an approximately 24-h period under non-physiological loading [3]. Predictions are consistent with studies showing that excessive mechanical loading disrupts BMAL1 oscillations [3], while circadian disruption impairs extracellular matrix homeostasis promoting IVD degeneration [4]. By integrating mechanical loading, nutrient transport, MT and CR regulation, the model provides a mechanistic framework linking altered mechanical environments to time-dependent Human activities.

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References

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